



3.6 Vectors and scalar/vector products (HL)

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A vector $\mathbf{v} = (v_1, v_2, v_3)$ has magnitude $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$. The scalar (dot) product $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$ gives the angle θ between vectors, and $\mathbf{a} \cdot \mathbf{b} = 0$ indicates perpendicular vectors. The vector (cross) product $\mathbf{a} \times \mathbf{b}$ is a vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, with magnitude $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta$, used to find areas of parallelograms/triangles and normals to planes.

Worked example: follow the reasoning

Given $\mathbf{a} = (2, 1, -2)$ and $\mathbf{b} = (1, 3, 0)$, find $\mathbf{a} \cdot \mathbf{b}$ and the angle between them.
 $\mathbf{a} \cdot \mathbf{b} = 2(1) + 1(3) + (-2)(0) = 2 + 3 + 0 = 5$.
 $|\mathbf{a}| = \sqrt{4 + 1 + 4} = 3$, $|\mathbf{b}| = \sqrt{1 + 9 + 0} = \sqrt{10}$.
 $\cos\theta = 5/(3\sqrt{10}) = 5/9.487 = 0.5271$, $\theta = \arccos(0.5271) = 58.2^\circ$ (3 s.f.)

Quick check · multiple choice

Vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy $\mathbf{a} \cdot \mathbf{b} = 0$ (both nonzero). What does this tell us about $\mathbf{a}$ and $\mathbf{b}$?

- A They are parallel
- B They are equal
- C They are perpendicular
- D They have equal magnitude

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Let $\mathbf{a} = (1, 2, -1)$ and $\mathbf{b} = (3, 0, 2)$. [7]
- (a) Find $\mathbf{a} \times \mathbf{b}$.
 - (b) Verify that $\mathbf{a} \times \mathbf{b}$ is perpendicular to $\mathbf{a}$ by computing $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a}$.
 - (c) Find the area of the triangle with two sides given by vectors $\mathbf{a}$ and $\mathbf{b}$.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $\mathbf{a} \times \mathbf{b} = (2(2) - (-1)(0), (-1)(3) - 1(2), 1(0) - 2(3)) = (4 - 0, -3 - 2, 0 - 6) = (4, -5, -6)$.

(b) $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 4(1) + (-5)(2) + (-6)(-1) = 4 - 10 + 6 = 0$. Confirmed perpendicular.

(c) $|\mathbf{a} \times \mathbf{b}| = \sqrt{16 + 25 + 36} = \sqrt{77}$. Area of triangle $= (1/2)|\mathbf{a} \times \mathbf{b}| = \sqrt{77}/2 \approx 4.39$.

Quick check answer

C. A zero dot product between nonzero vectors means the angle between them is 90° , i.e. perpendicular.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.